Term Project: Milestone 1
11 April 2026

Project Subject Area:

This project will analyze the relationship between a movie's production budget and its reception by audiences and critics by analyzing financial performance metrics, ratings, and audience sentiment. By joining the various data sources, I will be able to determine if a higher budget translates to higher scores from critics or if independent films achieve a better return on investment.

Data Sources:

Flat File: TMDB 5000 Movie Dataset
This is from Kaggle where they were positing if companies (like Pixar) had found a formula for movie success. They were looking to see if they could predict which films would be highly rated or a commercial success. This will be my base dataset as it includes the title, budget, production company, revenue, title, and some critic metrics. This csv is pandas ready to be loaded into a dataFrame.
Link or Flat File uploaded: https://www.kaggle.com/datasets/tmdb/tmdb-movie-metadata?resource=download

API: OMDb API
This is a RESTful web service that returns movie information including content and posters for movies. It includes IMDB id, movie title, the plot as well as the type if it's a movie, series, or episode. This also provides rotten tomato scores, ratings, votes and award information. I will use the requests library to send a GET request for each movie title, processing the resulting JSON objects to extract ratings.
Link : https://www.omdbapi.com/
I have already applied for, and received my token for API connection use, in an email.

Website: Box Office Mojo
This is a website by IMDBPro that includes data tables for several different tables of box office records including domestic, international, worldwide, and all time. There is an "Indices" tab that shows by the Brands, Franchises, or Genres the total gross profits along with the number of releases, including the number 1 release for each Brand along with its lifetime gross profit. This site also includes theater counts for top performing films along with current lifetime gross earnings. I will use the requests to fetch the page content and BeautifulSoup to parse the HTML <table> tags directly into a structured format.
Link: https://www.boxofficemojo.com/chart/top_lifetime_gross/

Relationships

To create the unified dataset, I will be able to use Movie Titles as the connection point between the different datasets since each dataset includes the movie title. I will use pandas to standardize titles by removing extraneous whitespace or special characters, and convert to all upper or lower case across all the sources. Once standardized, I will be able to do a merge in pandas using the title as the index to create the master dataset. Both the CSV file from Kaggle and the Website, Box Office Mojo, both contain a list of movies and financial data, so I will need to validate the entries across these sources to see if they match.

Project approach/plan

I will use Jupyter Notebook for my analysis to allow my code, analysis, and visualizations to all live in one place. I will load the CSV first and handle missing values and identify any outliers. Then I will use requests and BeautifulSoup to target the HTML table rows and cells from Box Office Mojo and convert the web data into a DataFrame. After this I will be able to write a loop using the requests library to query the OMDb API. Then I will merge all three DataFrames using pd.merge() and then use Matplotlib and Seaborn to create visualizations.

What concerns/challenges you think you will face with the data/project topic

Challenges could include having issues of matching titles in case they list them differently on each source, which could affect the consistency of the data. I may need to do fuzzy matching or string manipulation to ensure titles match correctly. The CSV contains 4,804 records, so I have a very large dataset here. We only require 1,000 rows, so I could cut down on the data but I think if I did cut down on the dataset that it would be difficult to decide which records I would keep. If I deleted too many rows that were not created by large production companies, then I would not be accurately representing the data I was able to obtain. I also notice that I see some $0 revenue entries in the CSV, so I need to decide how I will handle these. Were these released to TV movies that cannot directly attribute revenue made? So I feel like I will need to do some analysis on these $0 entries to decide how to best handle them.

Ethical Implications of your project topic

To ensure data integrity when I'm cleaning the data, I need to be careful not to delete records that have '0' budget or revenue or very low budgets or revenue, in case they represent indie films or made for TV movies and not just bad data. To ensure total transparency, I will document any assumptions I make during my analysis. In case a movie is not found in the API, I will exclude it from the final analysis rather than inserting placeholder values. I will also need to make sure that I verify the Box Office Mojo to ensure that I am respectful to their servers while I do my scraping. I will also make sure to strictly adhere to the OMDb API attribution guidelines for usage. For my API key for OMDb API I have a 1,000 daily limit for calls so I will be sure to do multiple pulls over multiple days if I need to increase my dataset size.